

Qatar-1b

R-2549

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_190716 · created 2026-08-26T00:46:02+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458680.803 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.031
Transit depth (Rp/Rs)^2	3.7 +/- 1.19 [%]
Orbital Inclination (inc)	82.27 +/- 0.98
Airmass coefficient 1 (a1)	0.683 +/- 0.021
Airmass coefficient 2 (a2)	0.31 +/- 0.12
Scatter in the residuals of the lightcurve fit is	3.11 %
Best Comparison Star	#2 - [376, 272]
Optimal Aperture	1.95
Optimal Annulus	5.57
Transit Duration (day)	0.0605 +/- 0.0097

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.031 vs 0.1463 (deviation 1.09σ)
Duration fit vs published	0.0605 d vs 0.0692 d (deviation 0.66σ)
Mid-time O–C	-3.2 min
Depth SNR	4.6
Residual scatter	3.11 %

Light curve

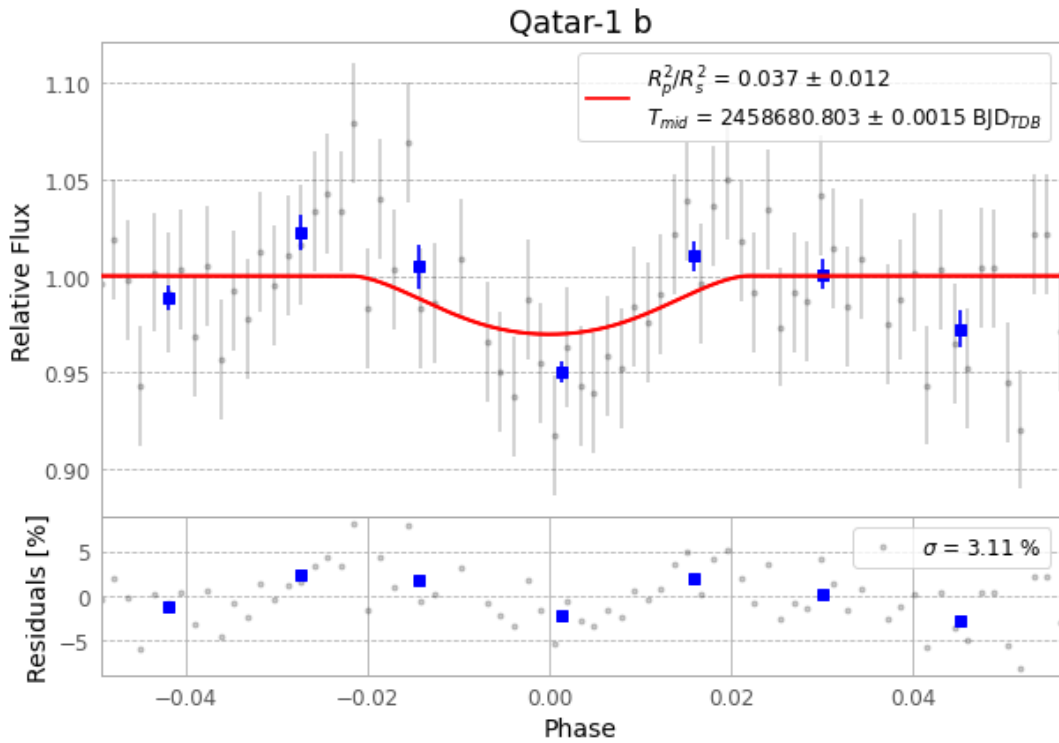


Figure 1 — FinalLightCurve_Qatar-1 b_2019-07-15.png (SHA-256 f71e2c1d5de1b81d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [446, 249], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4549602318a1d294...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

73 FITS frames (48 MB), Qatar-1190716053312.FITS → Qatar-1190716090912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-190716.log (d7e75c1f32ac...), FinalLightCurve_Qatar-1 b_2019-07-15.png (f71e2c1d5de1...), FinalLightCurve_Qatar-1 b_2019-07-15.csv (dd94ff0c026b...)

Compute resources

Wall time 153.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 00:46Z.